

2. grid method.

① pre-smoothing by DGS.

② get residual. for u, v, p .

$$r_u = f_1 + \Delta u - p_x$$

$$r_v = f_2 + \Delta v - p_y$$

$$r_p = g - \nabla_x u - \nabla_y v$$

in details:

$$i = 2:n-1$$

$$j = 2:n$$

$$r_u(i,j) = f_1(i,j) - \left(4u(i,j) - u \begin{pmatrix} i+1,j \\ i-1,j \\ i,j+1 \\ i,j-1 \end{pmatrix} \right) / h^2$$

$$- (p(i,j) - p(i,j-1)) / h$$

$$r_v(j,i) = f_2(j,i) - \left(4v(j,i) - v \begin{pmatrix} j+1,i \\ j-1,i \\ j,i+1 \\ j,i-1 \end{pmatrix} \right) / h^2$$

$$- (p(j-1,i) - p(j,i)) / h$$

i, j run

$$r_p(i,j) = g(i,j) - \underbrace{(u(i,j+1) - u(i,j)) / h}_{\text{div}} - (v(i,j) - v(i+1,j)) / h$$

② restriction

$u, v \rightarrow 6$ points

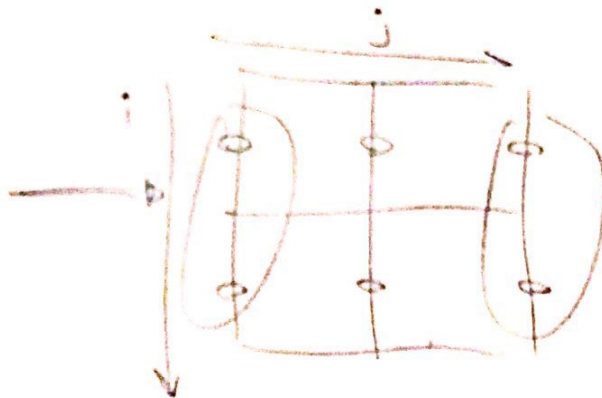
$p \rightarrow 4$ points



$u_f: 4 \times 5$

$v_f: 3 \times 4$

$p_f: 4 \times 4$



boundary
use u_D
 u_N

$u_c: 2 \times 3$

$v_c: 3 \times 2$

$p_c: 2 \times 2$

restriction of u

$2j-2 \quad 2j-1 \quad 2j$

$\downarrow \quad \downarrow \quad \downarrow$

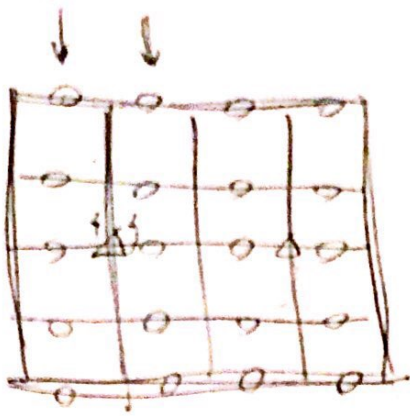


$i: 1 \sim n_c$

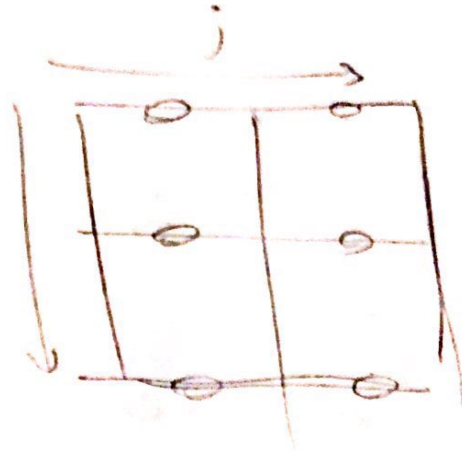
$j: 1 \sim n_c+1 \rightarrow \text{interior } 2 \sim n_c$

restriction for v

z_{j-1} z_j



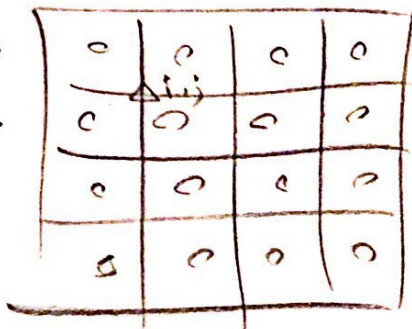
$\rightarrow$



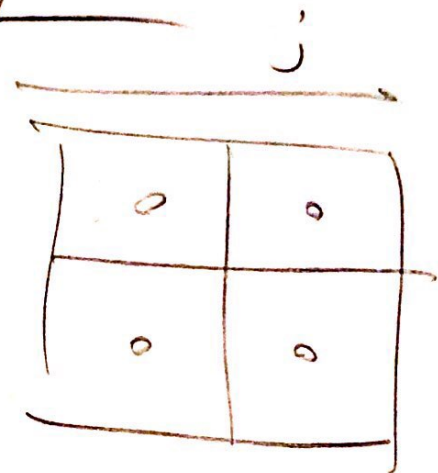
$i : 1 \sim nc+1 \xrightarrow{\text{interior}} 2 \sim nc$
 $j : 1 \sim nc$

restriction for w

z_{j-1} z_j

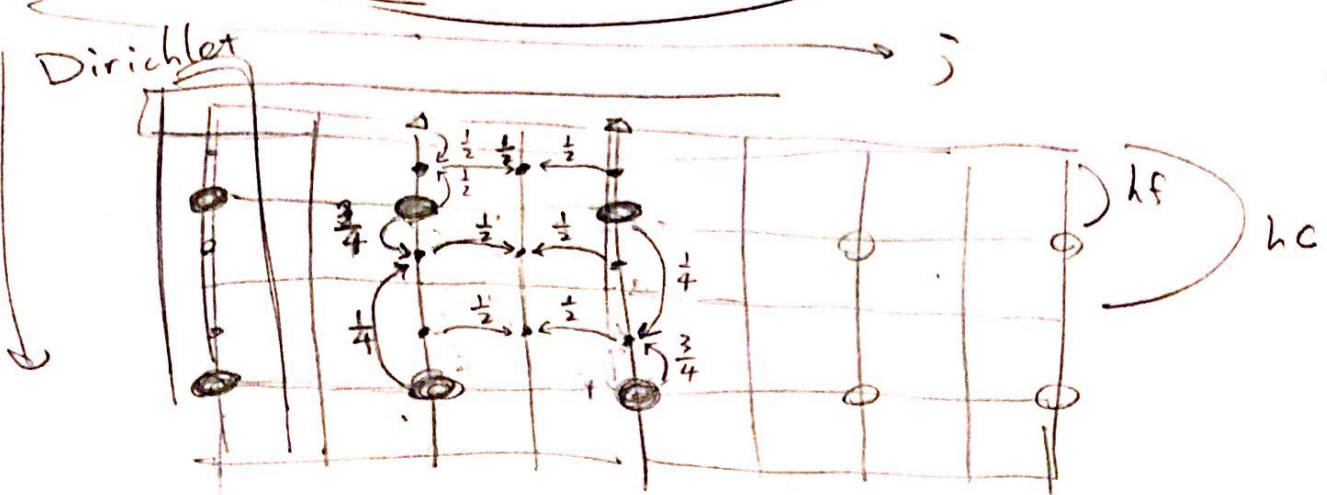


$\rightarrow$

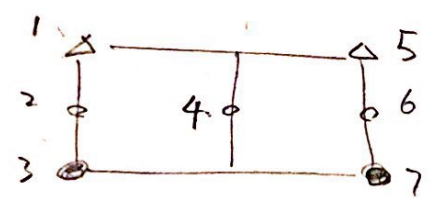


$i : 1 \sim nc$
 $j : 1 \sim nc$

prolongation of u



points near boundary :



u_1, u_5 known by Dirichlet boundary
 u_3, u_7 known by coarse grid

$$u_2 = \frac{1}{2} (u_1 + u_3)$$

$$u_6 = \frac{1}{2} (u_5 + u_7)$$

then $u_4 = \frac{1}{2} (u_2 + u_6)$

interior u

Top :

int $u(i-1, j)$
 $u(i-2, j-1)$
 $u(i-1, j-1)$
 $u(i, j)$

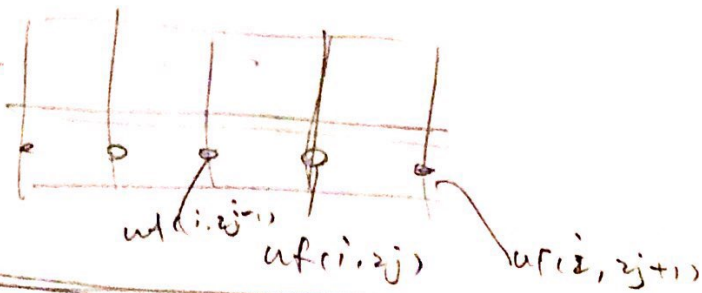
i :

$$u((i-1) \times 2 + 1, j-1)$$

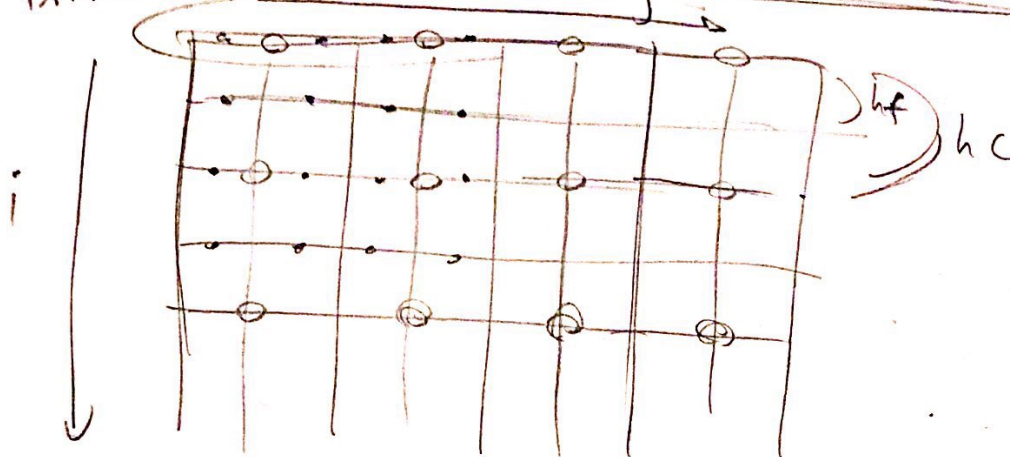
$$\sum_{i=1}^{i-1}$$

$$(j-1) \times 2 + 1$$

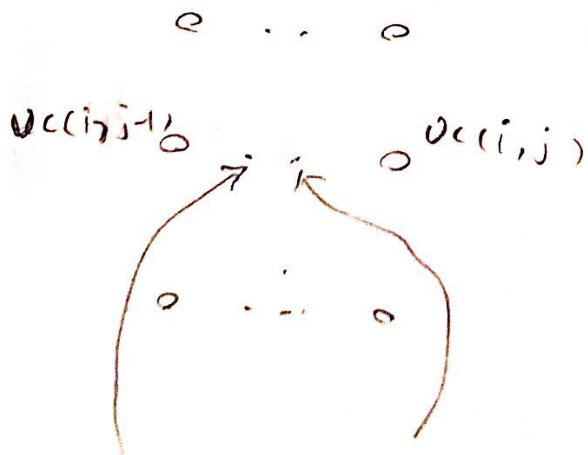
$$2j - 1$$



✓ Dirichlet



prolongation of u .



$u(2i-1, 2j-2)$ $u(2i-1, 2j-1)$



$(j-1)*2+1$

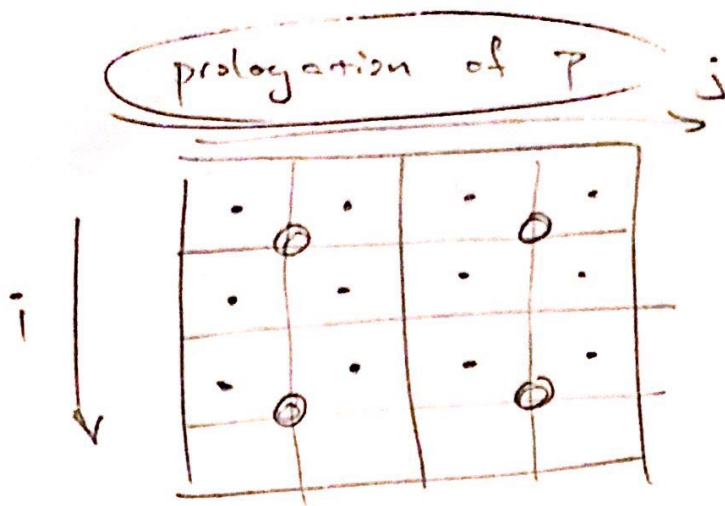
$2j-2+1$

$2j-1$

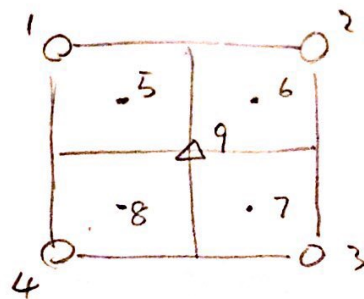
$(i-1)*2+1$

$2i-2+1$

$2i-1$



for interior :



p_1, p_2, p_3, p_4 known by coarse

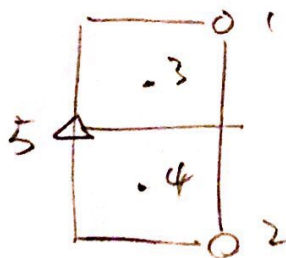
$$p_9 = \frac{1}{4} (p_1 + p_2 + p_3 + p_4)$$

$$p_5 = \frac{1}{2} (p_1 + p_9)$$

$$p_6 = \frac{1}{2} (p_2 + p_9)$$

etc.

for boundary :



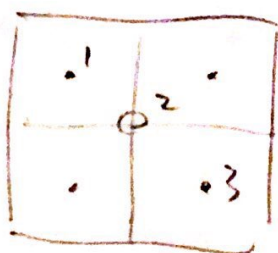
p_1, p_2 known by coarse

$$p_5 = \frac{1}{2} (p_1 + p_2)$$

$$p_3 = \frac{1}{2} (p_1 + p_5)$$

$$p_4 = \frac{1}{2} (p_2 + p_5)$$

for corner :



p_2 known by coarse

p_3 solved as interior point

$$\text{let } p_2 = \frac{1}{2} (p_1 + p_3)$$

$$p_1 = 2p_2 - p_3$$

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0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

$$\boxed{g} = \partial_{\vec{n}} u = 0.$$

Solve $\hat{A}_p e_p = r_p.$

bdd.

$$r_p = \frac{\textcircled{3} \textcircled{-1} \textcircled{-1} \textcircled{-1}}{h^2}$$

boundary.

corner

$$r_p = \frac{\textcircled{2} \textcircled{-1} \textcircled{-1}}{h^2}$$

Note. this also will be used in
the distribute correction (last step)